

Mathematics & Statistics
MATH 3CV3
Basic Complex Variables
Winter 2027



SCIENCE

Instructor Information



Gregory Cousins
Email: cousingd@mcmaster.ca
Office Hours:
TBD

Course Information

Math 3CV3: This is an introductory course on complex analysis, with a focus on computation with and manipulation of complex functions and series. The course begins with an introduction to complex numbers, their algebra and representations, and then moves on to complex functions and their properties. Extra focus will be given to analyticity of complex functions, zeros, poles, path integrals of analytic functions, and computations involving residues.

Comparison with Math 3CF3: This course is suitable for most students in a math and stats program. Math 3CF3 (Theory of Complex Functions) will move at a faster pace, be more proof intensive, and will focus on the structure and classification of analytic functions, with much less focus on computation. You are encouraged to register in Math 3CF3 if you are interested in theory, and especially if you intend to pursue graduate studies in mathematics or statistics.

Course Dates: 01/04/2027 - 04/07/2027

Units: 3.00

Course Delivery Mode: In Person

Course Description: Computational aspects of complex variables and applications. Complex numbers, polar exponential notation, the complex plane. Complex functions, limits, derivatives and analytic functions. Definite integrals, contour integrals, antiderivatives. Cauchy's Theorem. Cauchy's Integral Formula.

Liouville's Theorem. Convergence of series, geometric series, Taylor series, Laurent series, integration and differentiation of series. Residue theorem, poles and zeros, plus applications. Three lectures, one tutorial; one term Prerequisite(s): One of MATH 2MC3, 2XA3 (or MATH 2X03), or ISCI 2A18 A/B; and MATH 1C03 Not open to students with credit or registration in MATH 3CF3. Not open to students with credit in MATH 3X03.

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)
- [Open Education Resources](#)

Course Learning Outcomes

- To be able to analyse complex-valued functions of a complex variable in the same ways as in calculus for real-valued functions of a real variable. This includes being able to evaluate limits, derivatives, and contour integrals, as well as developing analytic functions into Taylor series.
- Understand the ways in which analytic functions are different from real differentiable functions: Cauchy-Riemann equations and consequences, regularity, unique continuation, properties of zero sets, maximum principle
- Understand the elementary properties of complex functions: poles, zeros, residues and simple applications to the evaluation of improper real integrals.
- To manipulate and compute with complex series, including finding Taylor and Laurent series expansions of complex functions.

Required Materials and Texts

Please sign in with your MacID [here](#) to view your booklist

Complex Variables and Applications

ISBN: 978-0-07-338317-0

Authors: J. W. Brown and R. V. Churchill

Publisher: McGraw-Hill

Publication Date: 2014

Edition: 9th

Optional Course Materials

Please sign in with your MacID [here](#) to view your booklist

The students' solution manual to the textbook

Class Format

In Person

Course Evaluation

Your course average will consist of the following components:

1. Five written homework assignments: 10% (2% each)
2. Two scheduled Midterm tests: 50% (25% each)
3. Registrar's office supervised final exam 40%

This breakdown is subject to change, and the instructor may consider alternative grading schemes at the end of the term.

Undergraduate Grading Scale

The breakdown of McMaster's 12-Point Grading Scale.

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62

Grade	Equivalent Grade Point	Equivalent Percentages
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Graduate Grading Scale

Graduate Student Grading Scale (Except for MBA and Master of Finance)

Grade	Points	Equivalent Percentage	Pass/Fail
A+	12	90-100	P+
A	11	85-89	P
A-	10	80-84	
B+	9	77-79	
B	8	73-76	
B-	7	70-72	
F	0	69 and under	F

MBA and Master of Finance Grading Scale

Grade	Points	Equivalent Percentage	Pass/Fail
A+	12	90-100	P+
A	11	85-89	P
A-	10	80-84	
B+	9	75-79	
B	8	70-74	
B-	7	60-69	
F	0	59 and under	F

Course Schedule

Tentative Lecture Schedule

The following schedule is tentative and may be adjusted depending on the pace of the class, cancellations, statutory holidays, and the timing of assessments.

Essential examinable material will be drawn primarily from Lectures 1–31.

Lectures 32–35 are flexible and may be used for additional applications, review, or enrichment if time permits.

Lecture 1: Complex numbers in rectangular form; algebraic operations, conjugates, modulus, and distance.

Lecture 2: Polar and exponential forms; arguments, principal arguments, multiplication, division, and powers.

Lecture 3: Roots of complex numbers and roots of unity.

Lecture 4: Regions in the complex plane; neighbourhoods, open and closed sets, boundaries, domains, and connectedness.

Lecture 5: Complex functions as mappings; images and preimages of regions under elementary transformations.

Lecture 6: Limits and continuity of complex functions.

Lecture 7: Complex differentiability and the usual rules of differentiation.

Lecture 8: The Cauchy–Riemann equations and sufficient conditions for differentiability.

Lecture 9: Analytic and harmonic functions; basic harmonic-conjugate calculations.

Lecture 10: The complex exponential function.

Lecture 11: Complex logarithms, branches, branch cuts, and arbitrary complex powers.

Lecture 12: Complex trigonometric and hyperbolic functions.

Lecture 13: Contours, parametrizations, orientation, and the definition of the contour integral.

Lecture 14: Direct computation of contour integrals.

Lecture 15: Estimates for contour integrals and the ML inequality.

Lecture 16: Antiderivatives and independence of path.

Lecture 17: The Cauchy–Goursat theorem; simply and multiply connected domains; deformation of paths.

Lecture 18: The Cauchy integral formula.

Lecture 19: Cauchy derivative formulas, Cauchy's inequalities, Liouville's theorem, and basic consequences.

Lecture 20: Complex sequences, series, and power series.

Lecture 21: Taylor series and disks of convergence.

Lecture 22: Computational methods for Taylor series.

Lecture 23: Negative powers and Laurent series.

Lecture 24: Laurent-series computations in different annuli and coefficient extraction.

Lecture 25: Isolated singular points, zeros, and poles.

Lecture 26: Residues and computational formulas for residues.

Lecture 27: Cauchy's residue theorem and contour-integral applications.

Lecture 28: Evaluation of rational improper integrals over the real line.

Lecture 29: Improper integrals arising from Fourier analysis.

Lecture 30: Definite trigonometric integrals using the unit-circle substitution.

Lecture 31: Synthesis and method selection; choosing among direct parametrization, antiderivatives, Cauchy theory, series, and residues.

Lecture 32: Indented contours and Cauchy principal values, if time permits.

Lecture 33: Further computational applications and mixed residue problems, if time permits.

Lecture 34: Optional enrichment on keyhole contours and integration along branch cuts, if time permits.

Lecture 35: Optional computational workshop on mixed residue computations and method selection, if time permits.

Tests, review lectures, and recovery days will be inserted into the schedule as required. Material from Lectures 32–35 will not be assumed unless it is explicitly announced in class

Late Assignments

Normally late assignments will only be accepted as a result of accommodations approved by the SAS. Please contact the instructor in case of other special circumstances. The standard duration for completion of an assignment is one week from the date on which the assignment is given. **If the due date is more than one week from the date the assignment is announced, the extra days are to be counted as accommodation for the assignment being late.**

Absences, Missed Work, Illness

Please follow the link in the Approved Advisory Statements section below pertaining to Missed Academic Term Work.

Once approved by the advisors in the Associate Dean's Office, the corresponding percentages will be distributed as follows:

- **Midterms:** weight will be added to the weight of the final exam,
- **Homework Assignments:** if fewer than three assignments are MSAF'd, the weight will be averaged across the remaining assignments. If three or more assignments are MSAF'd (such as in the case of extended illness) the student will be required to complete supplementary material for grading.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No

penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it "shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also state in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Course Specific Relief for Missed Academic Work Information

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

The percentage associated with the missed work will be transferred to the final exam once the official approval from the Associate Dean's office is received.

Checking Grades & Regrade Policy

Checking Your Grades

Your marks will be recorded on Avenue. It is your responsibility to check that all grades entered into Avenue are recorded properly. **Check only the actual number of marks obtained, not the percentages computed by Avenue.** All grade concerns and discrepancies must be reported to the Instructional Assistant within a week of receiving the grade.

Regrade Policy

This should be done within one week of getting back the marked test or assignment and BEFORE the final exam is written. If the TA or instructor considers the justification to be insufficient (e.g. simply wanting a higher or passing grade is insufficient), the assignment or test will not be re-graded. If the justification is considered sufficient, the entire assignment/test will be re-graded. Please be aware that this may result in an **increase, decrease or no change** to the original mark. There will be no alternative assignments that can be completed by students for the purpose of increasing their final grade.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or

other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.